

Tax Incentive Market Analysis: Meeting Discussion Points

Updated August 3, 2026 | Part A: original framing from July 26 | Part B: what changed while actually building it | Part C: decisions needed today

PART A WHERE WE STARTED (JULY 26)

A1. Quick wins already applied

- Rafi's four data points applied to the live model: furnace baseline CapEx (479 ILS/kW net), incremental maintenance (0 for all three technologies, conservative direction), VSD degradation (0), VSD hours (~5,000, his real compressor-run-hours ceiling).
- Electricity tariff corrected to the real High-Voltage TAOZ (time-of-use) average (39.54 agorot/kWh).
- Built the A–C investment-decision row and verdict per technology block, with conditional formatting, per Daniel's request.

A2. The deadweight discovery and the payback-threshold fix

Replicating the model independently in Python (LibreOffice cannot recalculate in this sandbox) surfaced that **every technology was already NPV-positive without the incentive** under the original 20-year NPV adoption rule. Stress-tested three ways (tariff fix, 25-year horizon, 50%-of-CapEx replacement cost) . Still deadweight everywhere.

Fix: real firms approve efficiency capex on short internal payback hurdles, not a 20-year social NPV. **Threshold set to 3 years, sourced to Rafi's own number** from his first conversation with Omri about this model.

Technology	Payback B	Payback C	Verdict
Heat pumps 40kW	0.48	0.47	Worth it anyway (deadweight)
Heat pumps 70kW	0.55	0.54	Worth it anyway (deadweight)
Chillers 100RT	3.60	2.36	Incentive flips it (real additionality)
Chillers 500RT	4.53	2.73	Incentive flips it (real additionality)
VSD 45kW	0.63	0.58	Worth it anyway (deadweight)
VSD 150kW	0.63	0.58	Worth it anyway (deadweight)

A3. What this reframes for the chapter

- **Only chillers carry the impact claim.** Multiplying HP/VSD per-unit results by a market size would measure a transition happening anyway, not incentive impact.
- **Effort allocation follows the split.** Chillers get rigorous market sizing (real impact claim). HP/VSD get a lighter, order-of-magnitude estimate (only feeds a deadweight/fiscal-cost figure, not an impact number).
- **A policy recommendation falls out of this.** Consider restricting the accelerated-depreciation multiplier to chillers specifically, rather than one uniform multiplier. Precedented by Cyprus's own differentiated-multiplier design (already in the international review).
- **Fiscal cost needs two numbers.** Total fiscal cost = tax-shield benefit (C–B) summed across every adopting unit, all technologies (the incentive does not discriminate on additionality). Cost-effectiveness (ILS/tCO₂, ILS/MWh) can only be honestly computed for chillers, where there is real additional abatement to divide by.
- **A real asymmetry vs. the grant chapter, worth naming directly.** The grant chapter explicitly avoided an additionality claim. The tax chapter, built this way, makes one, for chillers only.

A4. Original market-sizing plan

- **Chillers:** CBS non-residential construction starts (Daniel's own method) × a sourced RT/m² figure → installed cooling capacity added per year.
- **Heat pumps:** national energy balance industrial mazut/diesel use, bounded by the heat pump's low-temperature ceiling (~80–90°C).
- **VSD:** national industrial electricity × a compressed-air-share benchmark, narrowed to variable-load compressors only.
- **PRTR flagged as a lead** for narrowing the heat-pump/VSD addressable pools with a real Israeli anchor, instead of borrowed international benchmarks.

PART B WHAT CHANGED WHILE WE ACTUALLY BUILT IT (AUGUST 3)

B1. Chillers: built and sourced

STATUS: DONE, one open tension

CBS Table 7 construction-starts data (2021–2025) pulled directly. Rather than one blended RT/m² figure, each of the 7 chiller-relevant CBS categories got its own sourced sq-ft/ton figure (cfm Distributors, HVAC-ENG.com, Trane), weighted by that category's actual construction share each year.

Result: ~128,600 RT/year average installed chiller capacity from new non-residential construction, 2021–2025. Full sourcing and formulas live in *chiller-market-sizing.xlsx*.

Open tension: two categories (Other public buildings, Industry & storage) have no direct source and fall back on a general commercial proxy, which happens to make industry read as the *least* cooling-intensive category, counter to intuition about industrial process cooling. Whether that's actually wrong depends on a scope question: is this model's "chiller" technology comfort cooling only, or does it also cover industrial process cooling? One data point pulling toward comfort-only: Daniel already confirmed the model's 3,000 hrs/year assumption is good "especially for comfort chillers."

B2. Heat pumps: the sizing base is sourced, but it's the wrong kind of number

STATUS: real data pulled, structural gap found

Pulled Israel's real industrial mazut/diesel consumption directly from CBS's Energy Balance 2019 (the real *xlsx* behind the published table, not scraped from PDF text): 48,995 tons diesel + 131,167 tons mazut ≈ **2.07 TWh/year**, industrial sector.

The catch: this is a *stock* (total existing fuel burned across the current installed base), not the annual *flow* of new heat pump installations the tax model actually needs. Chillers avoid this problem because new construction is inherently a flow; industrial ovens/boilers are not tied to new buildings the way comfort chillers are.

B3. The PRTR trail

STATUS: investigated, real findings, no temperature data exists

- **EcoTraders literally co-authored the national PRTR annual report** (page 4 credits D.H.B Med Ltd. and EcoTraders Ltd. as the consulting firms). Worth checking internally whether anyone here retained facility-level working data from that engagement.
- The published report gives one real, usable anchor: **47% of all PRTR-reported mazut consumption is asphalt production**, a high-temperature use, clearly not heat-pump-addressable. That bounds the low-temperature-addressable mazut share at 53% at most.

- Confirmed directly (Omri reviewed actual facility filings): PRTR reports fuel/pollutant quantities against a facility's sector code, never process temperature. No dataset anywhere records what temperature a facility burns its fuel at. This was never going to be a data-lookup problem.

B4. The mindset shift: a proposed framework, not another data hunt

Since heat pumps and VSD are both deadweight (Section A3, not carrying an additionality claim), the bar for these numbers is order-of-magnitude, not precision. The right move is a transparent structural framework with the genuinely-unknown piece explicitly flagged as an engineering judgment call, rather than continuing to search for data that was never going to exist.

Step	Heat pumps	VSD
Stock (real data, sourced)	CBS industrial mazut + diesel: 2.07 TWh (2019)	National industrial electricity × ~10% compressed-air share (DOE/Radgen international benchmark)
Addressable share (ask Rafi)	Low-temp fraction: 53% mazut ceiling already established (asphalt exclusion); diesel has no ceiling yet	Variable-load fraction of compressed-air electricity (only variable-load compressors benefit from VSD)
÷ Lifetime (already have it)	Converts stock to annual replacement flow	Same conversion

Proposal: apply this same three-step structure to both technologies. It keeps the chapter honest: chillers carry real rigor because they carry the impact claim; HP/VSD carry an appropriately light, clearly-labeled-as-judgment order-of-magnitude estimate, consistent with how this chapter is already structured to differ from the grant chapter.

B5. A caveat the stock÷lifetime formula needs

Stock÷lifetime assumes a mature, stable equipment population turning over predictably. PRTR shows industrial mazut consumption fell **95% cumulatively since 2012** (15% in 2023 alone). That's a market in steep decline, not a stable one. When old mazut equipment retires in a shrinking market, much of it is likely being replaced by **gas conversion**, not sitting there as an open heat-pump opportunity. Worth asking whether Rafi should build this haircut directly into his low-temp-share estimate, or whether it should be netted out as a separate, explicit adjustment.

PART C DECISIONS NEEDED FROM DANIEL TODAY

- Confirm the payback-threshold adoption rule (3-year hurdle) replaces the 20-year NPV rule as the model's adoption logic.
- Confirm chillers are the only technology carrying an additionality/impact claim; HP/VSD are treated as deadweight throughout the chapter.
- Confirm the effort-allocation split (chillers rigorous, HP/VSD light-pass/order-of-magnitude) for the market-sizing sections.
- Decide whether to recommend restricting the accelerated-depreciation multiplier to chillers only (Cyprus-style differentiated multiplier) as a chapter policy recommendation.
- Confirm the dual fiscal-cost reporting structure (total tax-shield cost across all technologies vs. chillers-only cost-effectiveness ratio).
- Confirm the chiller sizing method and result (~128,600 RT/year, CBS construction data, weighted RT/m² by category) is ready to carry into the chapter.

- **NEW (chiller scope):** is the model's chiller technology comfort cooling only, or does it also cover industrial process cooling? Resolves the Industry & storage proxy tension directly.
- **NEW (heat pump framework):** confirm the $\text{Stock} \times \text{Addressable Share} \div \text{Lifetime}$ structure as the right way to convert the CBS mazut/diesel baseline into an annual installation estimate.
- **NEW (heat pump low-temp share):** get Rafi's estimate for the low-temperature-addressable share of the remaining 53% of industrial mazut (post-asphalt-exclusion), plus a separate estimate for diesel (no ceiling established yet).
- **NEW (gas-conversion haircut):** get Rafi's read on how much of the declining mazut stock is converting to gas vs. representing a genuine heat-pump-replacement opportunity.
- **NEW (VSD framework):** apply the same $\text{Stock} \times \text{Addressable Share} \div \text{Lifetime}$ structure; confirm the 10% compressed-air-share benchmark (international, not Israel-specific) or ask if Rafi has a better anchor.
- **NEW (PRTR data year):** the CBS energy balance used is the 2019 edition (most recent found externally). Flag if Daniel/Rafi have access to a more current release, and whether anyone retained facility-level data from the PRTR report EcoTraders co-authored.

Part A reconstructed from the July 26 working capture (brainstorms/2026-07-26_payback-threshold-and-meeting-prep.md) since the original PDF file itself was not retained between sessions. Content is faithful to that capture, wording may differ slightly from the version originally delivered.